ELEC347 L9 20/11/25 at 9:00

Q4

Design a digital bandpass filter meeting the following specifications:

Centre frequency $F = 250\,Hz$

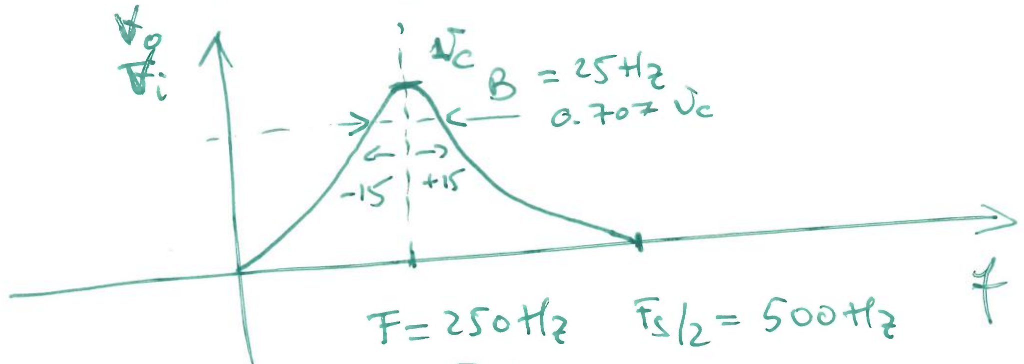
Filter order 2

Sampling frequency $F_S = 1000\,Hz$

Bandwidth $B = 25\,Hz = \pm 12.5\,Hz$

Solution

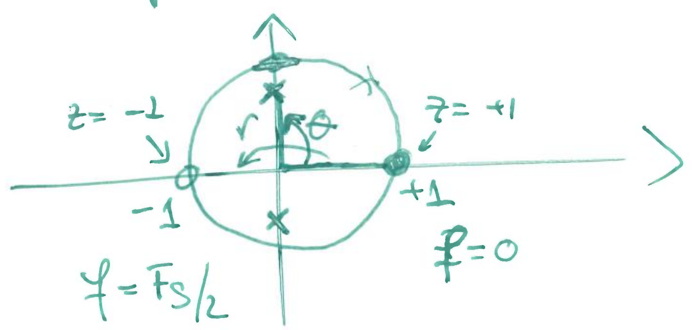
Frequency characteristic



$F = 250\,Hz = F_S/4$ $\quad F_S/2 = 500\,Hz$

Filter order 2 ⟶ Two poles (complex conjugate)
Two zeros.

Pole - zero placement

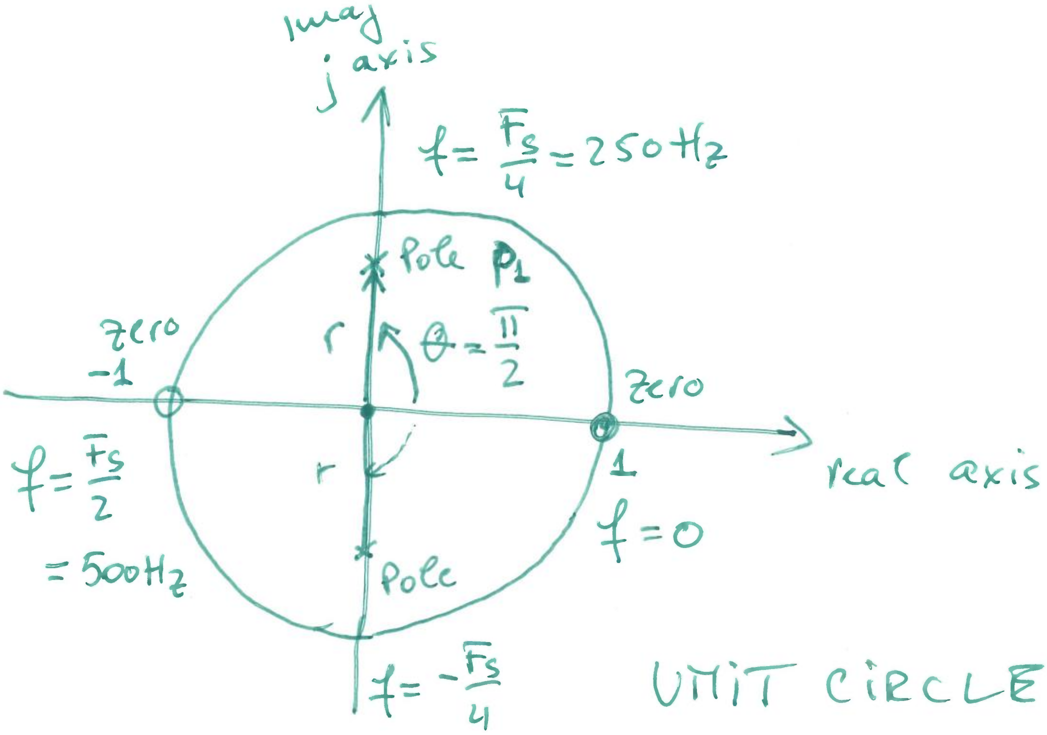


$$\theta = 2\pi \frac{F_S}{2F_S} = \pi$$

Poles at $re^{\pm j\theta}$

$$F = \frac{F_S}{4}$$

$$\Rightarrow \theta = 2\pi \frac{F}{F_S} = \frac{2\pi}{4} = \frac{\pi}{2}$$



$$p_1 = re^{j\theta} \qquad , \quad p_2 = re^{-j\theta}$$

$$= re^{j\frac{\pi}{2}} \qquad\qquad = re^{j-\frac{\pi}{2}}$$

$$= jr \qquad\qquad = -jr$$

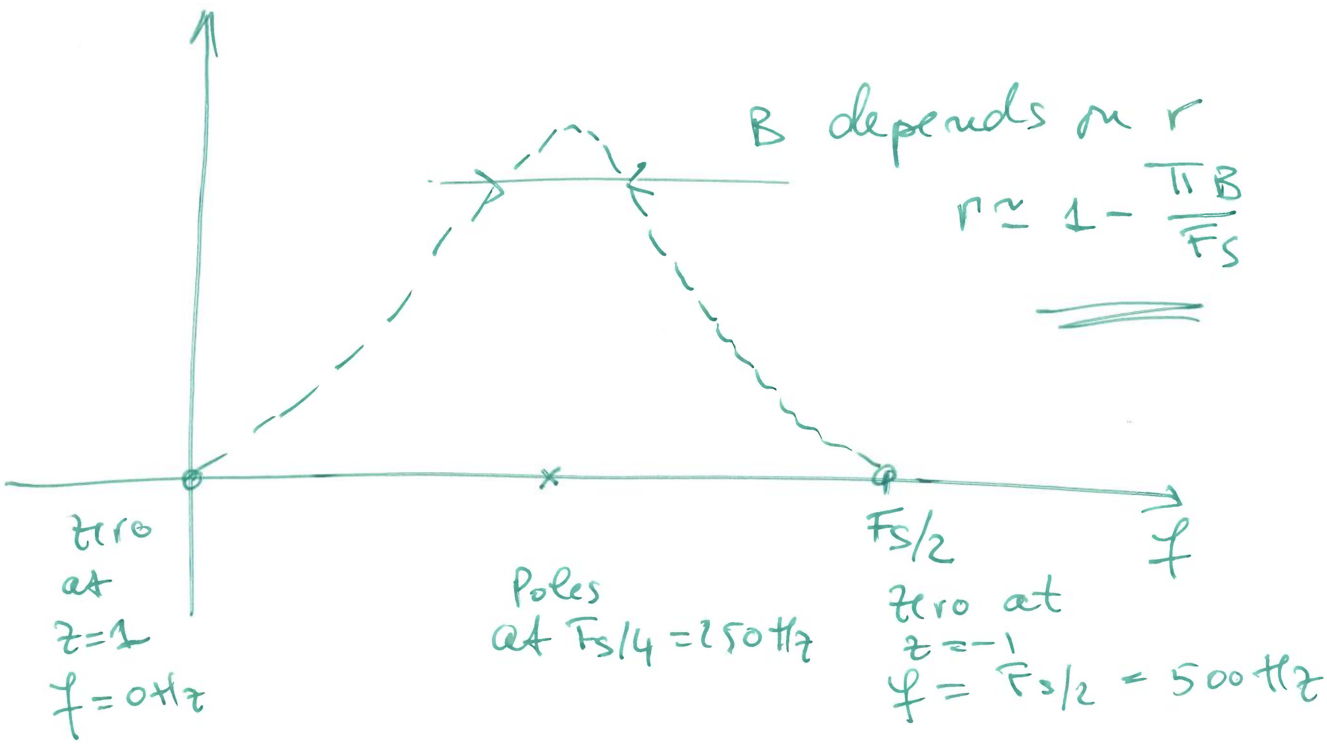
Euler

$$e^{j\theta} = \cos\theta + j\sin\theta$$

$$e^{j\frac{\pi}{2}} = \cos\frac{\pi}{2} + j\sin\frac{\pi}{2}$$
$$= 0 + j \times 1$$

$\Downarrow$

in frequency



$$r = 1 - \frac{\pi B}{F_s} = 1 - \frac{\pi \times 25}{1000}$$

$$= 1 - 0.07854 = 0.9214$$

$z=1$ $z=-1$

$$H(z) = \frac{(z-1)(z+1)}{(z - re^{j\theta})(z - re^{-j\theta})}$$

$$= \frac{(z-1)(z+1)}{(z-jr)(z+jr)} = \frac{z^2 - \cancel{z} + \cancel{z} - 1}{z^2 \cancel{+ jrz - jrz} - (jr)^2}$$

$$= \frac{z^2-1}{z^2-(jr)^2} = \frac{z^2-1}{z^2-j^2r^2} = \frac{z^2-1}{z^2+r^2}$$

$$= \frac{z^2-1}{z^2+0.849}$$

z-transfrm

Filter coefficients

$$\frac{z^2+0\times z-1}{z^2+0\times z+0.849} \begin{matrix} /z^2 \\ /z^2 \end{matrix} = \frac{1+0\times z^{-1} - z^{-2}}{1+0\times z^{-1}+0.849\times z^{-2}}$$

Difference quation

$$y(n) + 0.849\, y(n-2) = x(n) - x(n-2)$$

$a = [1 \;\; 0 \;\; 0.849] * M$ and truncated

$b = [1 \;\; 0 \;\; -1] * M$

$y = (y(n) \;\; y(n-1) \;\; y(n-2)]$

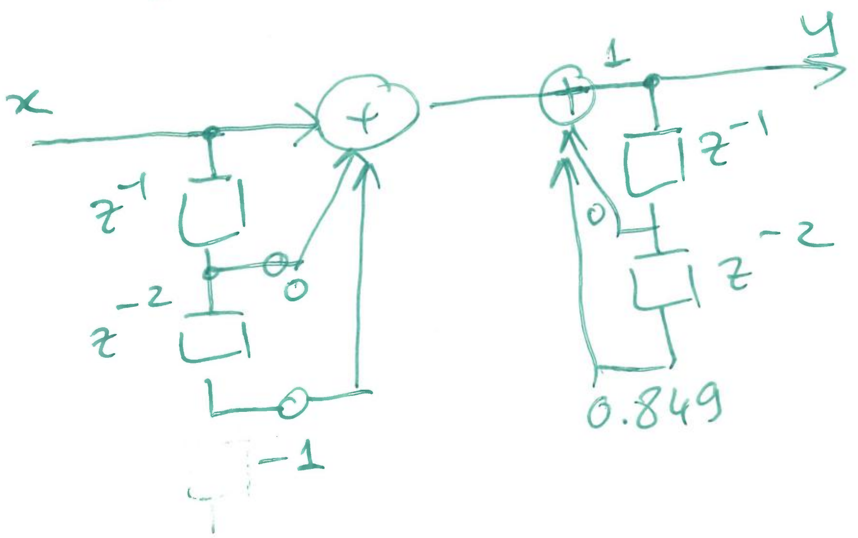
$x = \{x(n) \;\; x(n-1) \;\; x(n-2)]$

```
sum = 0
for (i=0; i<3; i++) {
    sum += y(i) a[i];
}
for (i=0; i<3; i++) {
    sum += x(i) b(i);
}
```

Convolution between filter coeff and data.

HW

Direct form



Q

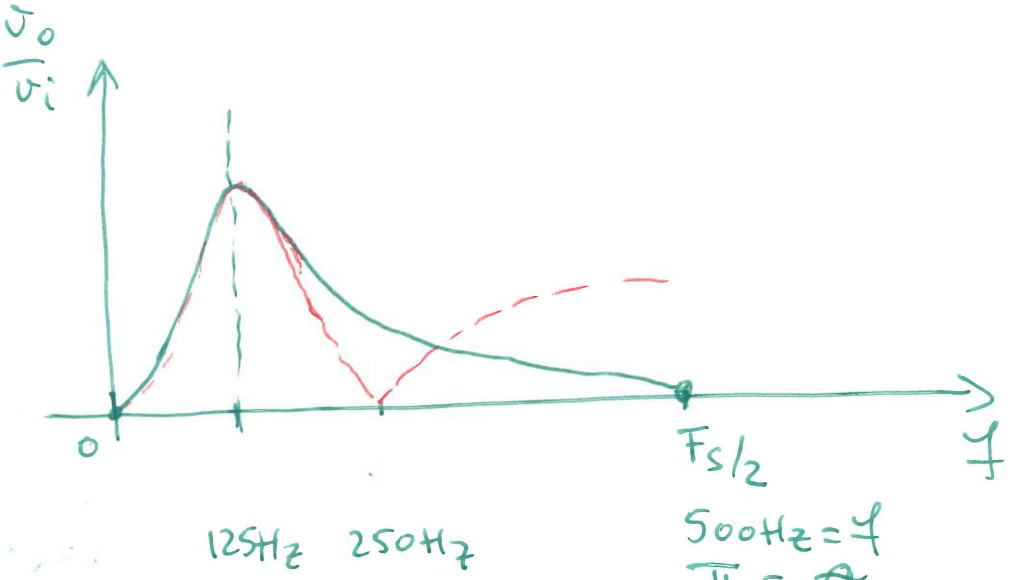
Design a bandpass filter with parameters

$F = 125\ Hz$

Order 2

$F_s = 1000\ Hz$

$B = 25\ Hz$

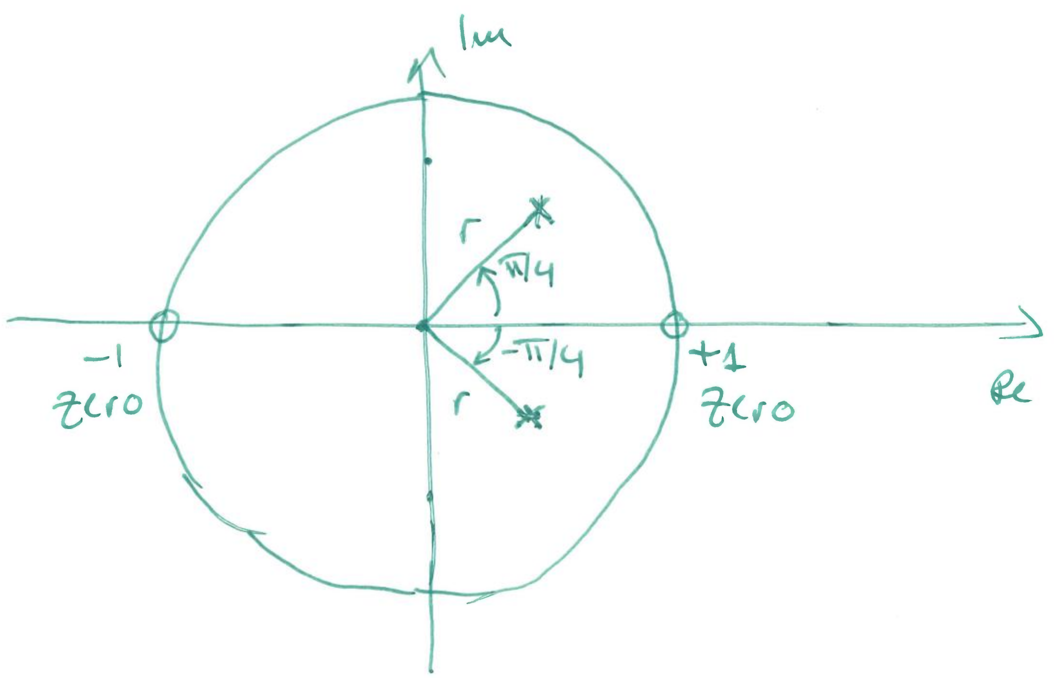


$\theta = 0$

$\theta = 2\pi \frac{125}{1000}$

$\theta = 2\pi \frac{f}{F_s}$

$= 2\frac{\pi}{8} = \frac{\pi}{4} = 0.25\pi$



$$r = 1 - \frac{\pi B}{F_s} = 1 - \frac{\pi 25}{1000} = 1 - 0.07854$$

$$= 0.9214$$

Zeros

$z = +1 \quad , z = -1$

Poles

$$z = re^{j\theta} = 0.9214\, e^{j\frac{\pi}{4}}$$

$$z = re^{-j\theta} = 0.9214\, e^{-j\frac{\pi}{4}}$$

$$H(z) = \frac{(z-1)(z+1)}{(z - re^{+j\theta})(z - re^{-j\theta})}$$

$$= \frac{z^2 - 1}{z^2 - re^{j\theta}z - re^{-j\theta}z + r^2 \underbrace{e^{j\theta - j\theta}}_{=1}}$$

$$= \frac{z^2 - 1}{z^2 - rz(e^{j\theta} + e^{-j\theta}) + r^2}$$

$$e^{j\theta} = \cos\theta + j\sin\theta$$

$$e^{-j\theta} = \cos(-\theta) + j\sin(-\theta)$$

$$e^{j\theta} + e^{-j\theta} = 2\cos\theta$$

$$\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

$$\frac{z^2-1}{z^2-2r\cos\theta\, z+r^2} = H(z)$$

$$r = 0.9214$$

$$\cos\theta = \cos\frac{\pi}{4} = \frac{1}{\sqrt{2}} = 0.707$$

$$H(z) = \frac{z^2-1}{z^2-2\times 0.707\times 0.9214 + (0.9214)^2}$$

$$H(z) = \frac{z^2-1}{z^2-1.414\times 0.9214 z + 0.849} \quad | z^{-2}$$

$$= \frac{z^2-1}{z^2-1.302z+0.849} \quad | z^{-2}$$

$$= \frac{1-z^{-2}}{1-1.302z^{-1}+0.849z^{-2}}$$

$$y(n) - 1.302\, y(n-1) + 0.849\, y(n-2)$$

$$= x(n) - x(n-2)$$